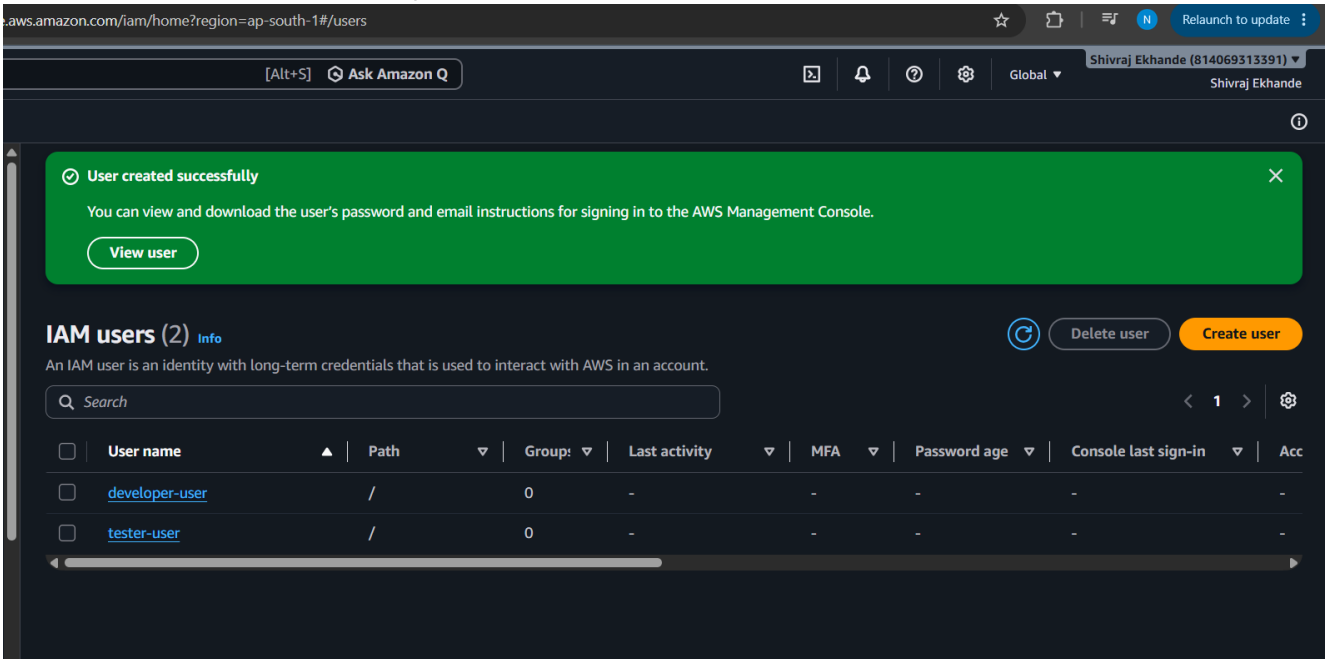
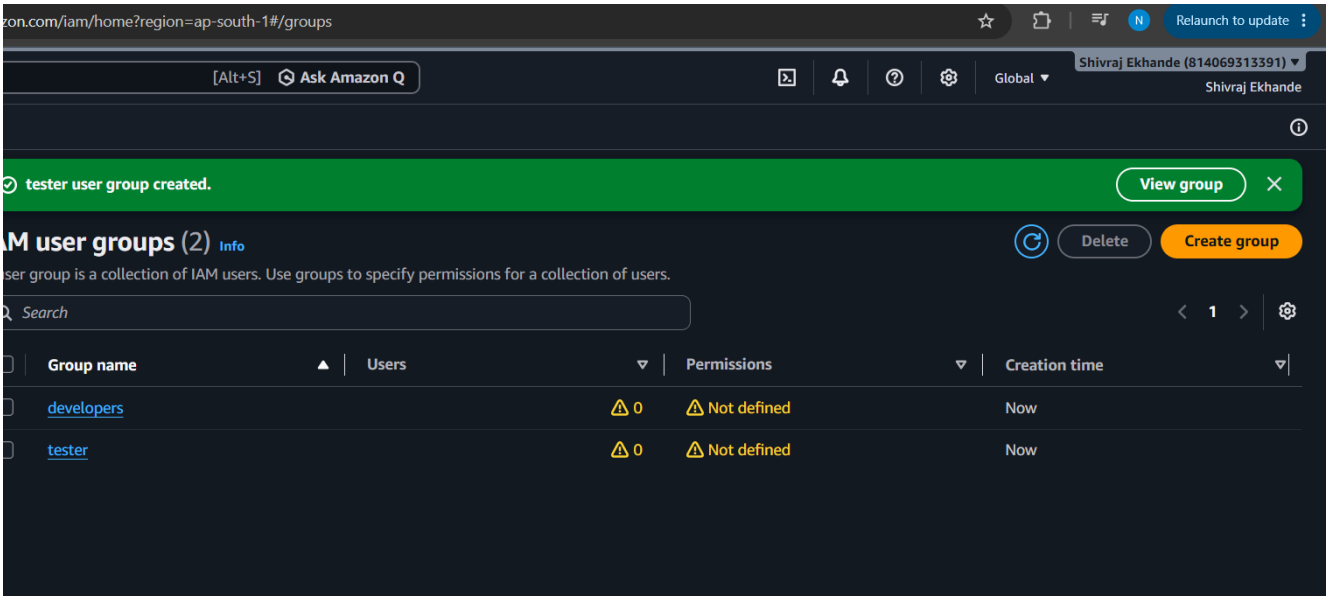
Section A — AWS Identity and Access Management (IAM)

Task 1: Create IAM users and groups

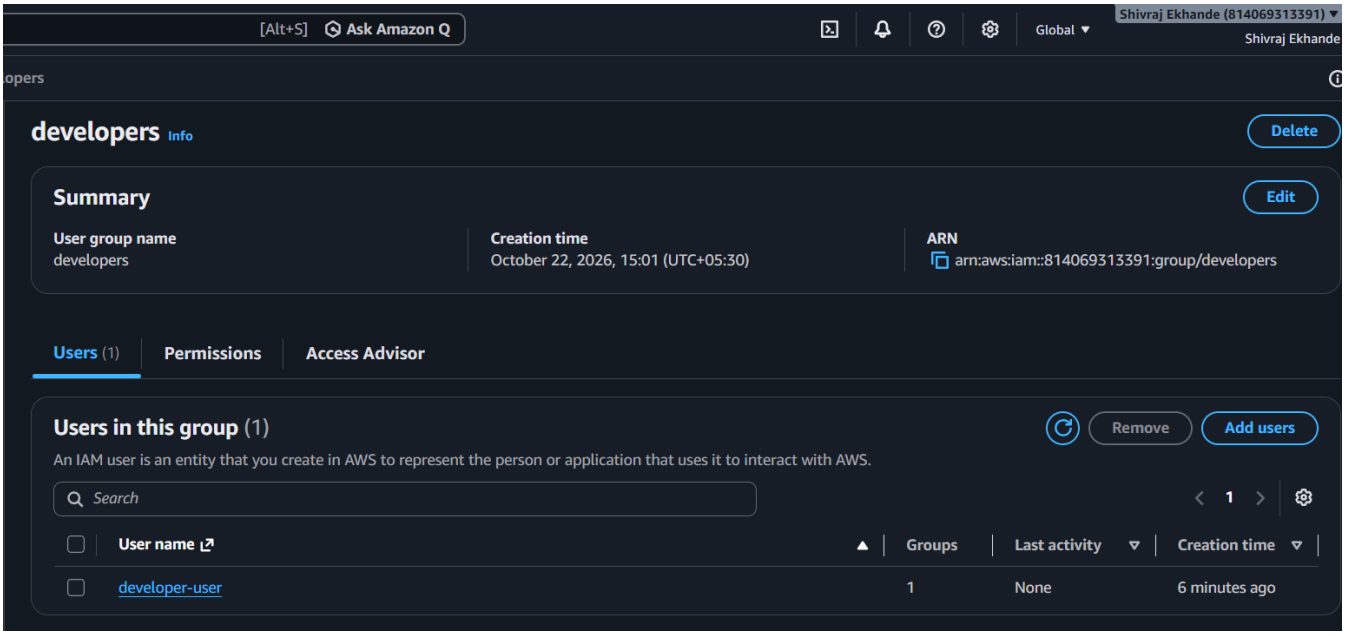
1. Create two IAM users named developer-user and tester-user.

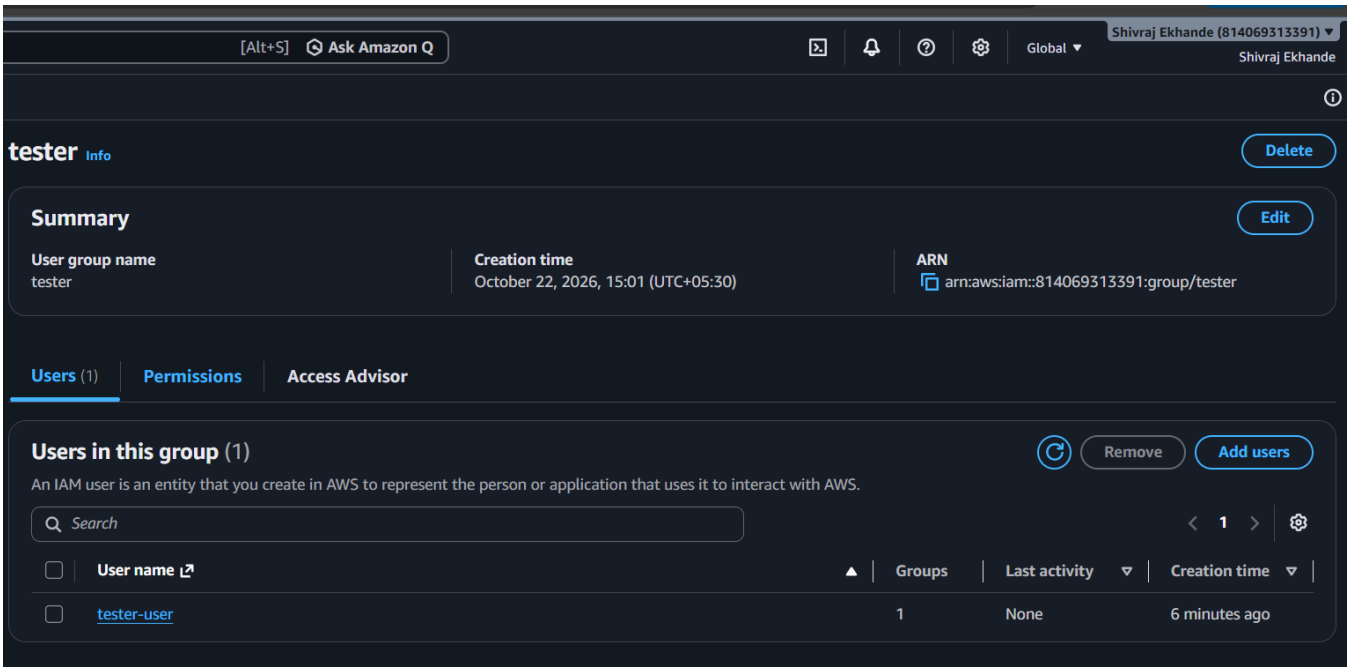


2. Create two IAM groups named Developers and Testers.



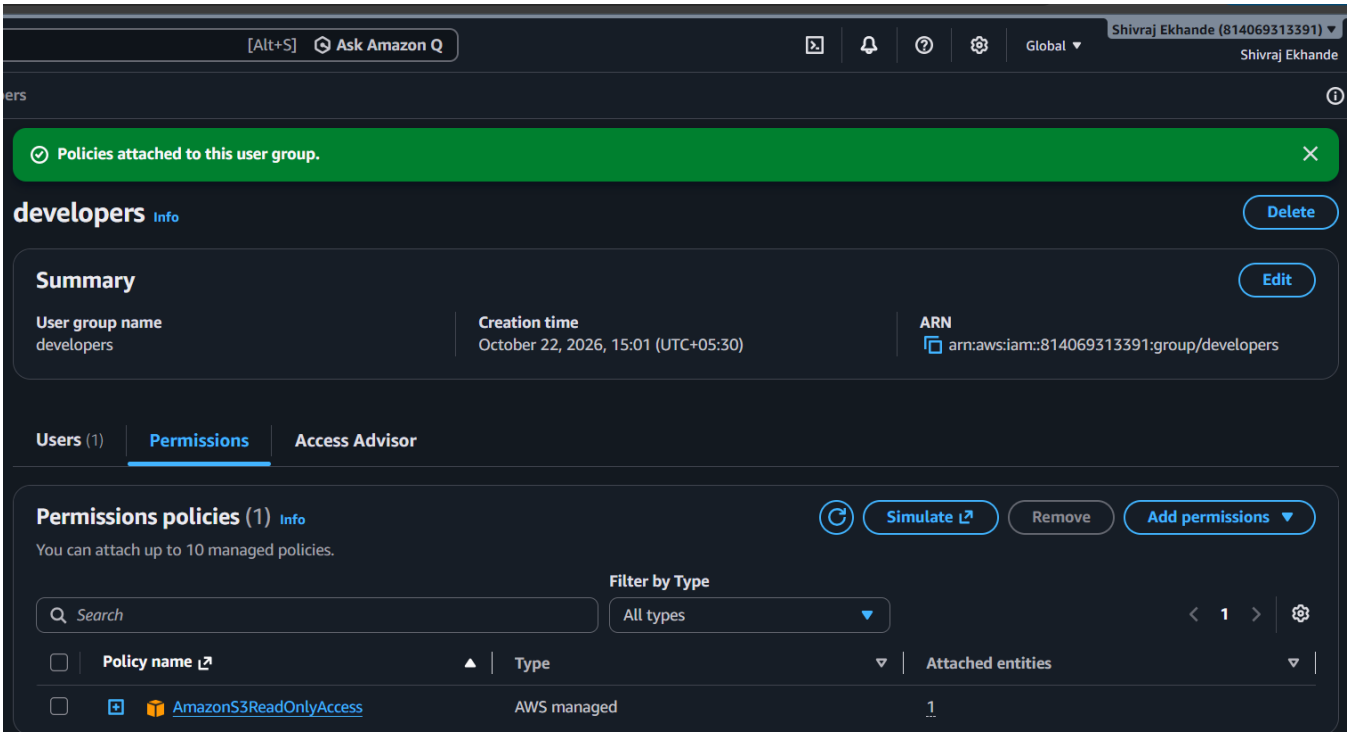
3. Add developer-user to Developers and tester-user to Testers.



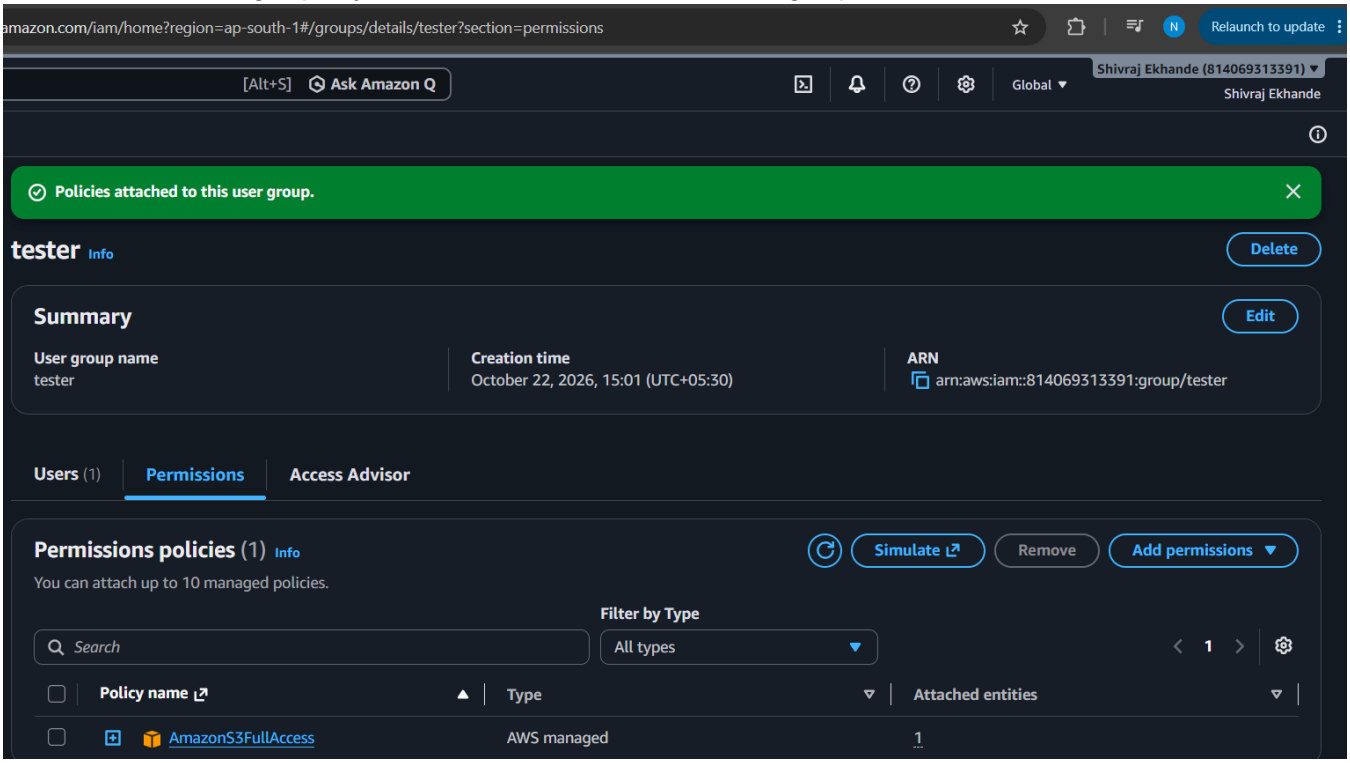


Task 2: Attach IAM policies

4. Attach the AWS managed policy AmazonS3ReadOnlyAccess to the Developers group.



5. Attach the AWS managed policy AmazonS3FullAccess to the Testers group.



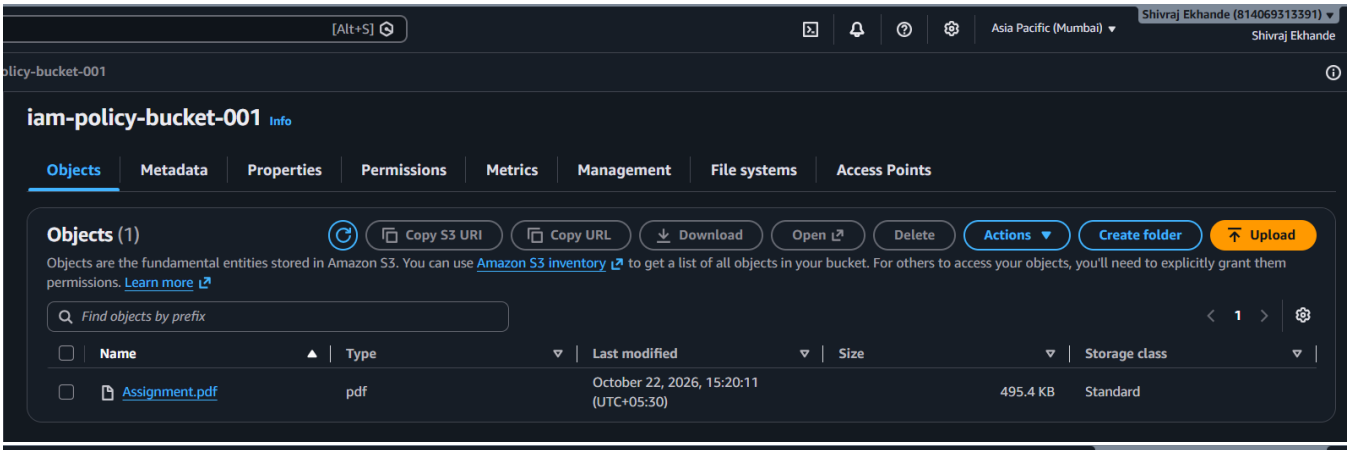
6. Write one or two sentences explaining the difference between these policies.
AmazonS3ReadOnlyAccess: Allows users to view and read S3 buckets and objects, but does not grant permission to upload or delete objects.
AmazonS3FullAccess: Allows users to perform a broad range of S3 actions, including creating buckets and uploading, reading, and deleting objects, subject to other applicable access controls.

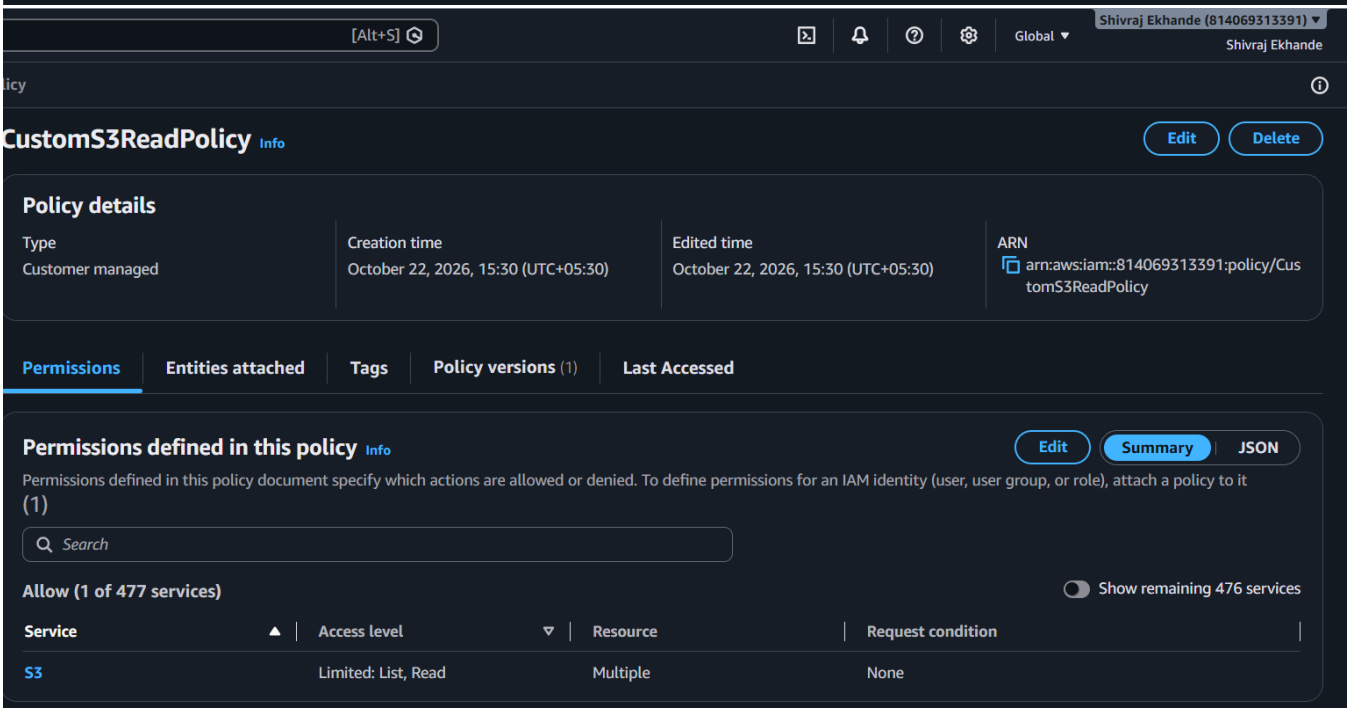
Task 3: Create a custom IAM policy

Create a customer-managed policy named CustomS3ReadPolicy.

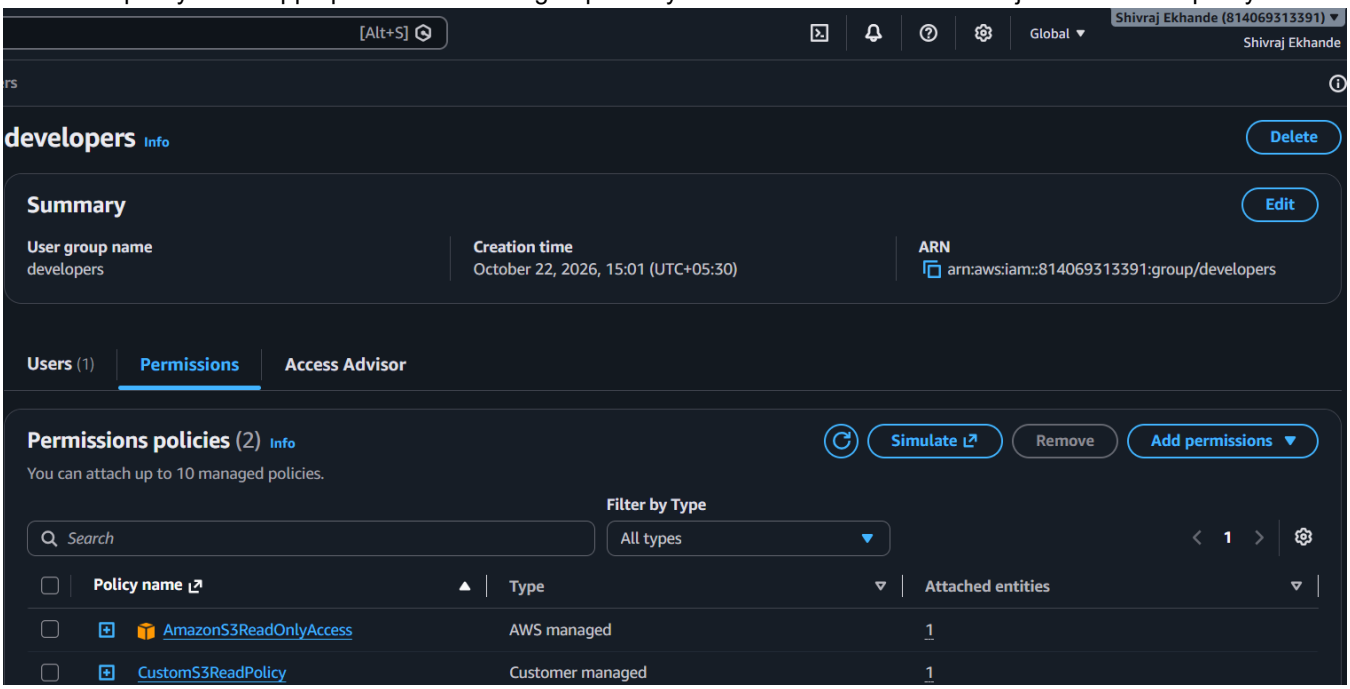
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket"
      ],
      "Resource": "arn:aws:s3:::iam-policy-bucket-001"
    },
    {
      "Sid": "Statement2",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::iam-policy-bucket-001/Assignment.pdf"
}
 ]
}
```

7. The policy must allow listing your practical S3 bucket and reading objects in that bucket only.


[Alt+S]
Asia Pacific (Mumbai)
Shivraj Ekhande (814069313391)
Shivraj Ekhande
iam-policy-bucket-001 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
Assignment.pdf
pdf
October 22, 2026, 15:20:11 (UTC+05:30)
495.4 KB
Standard

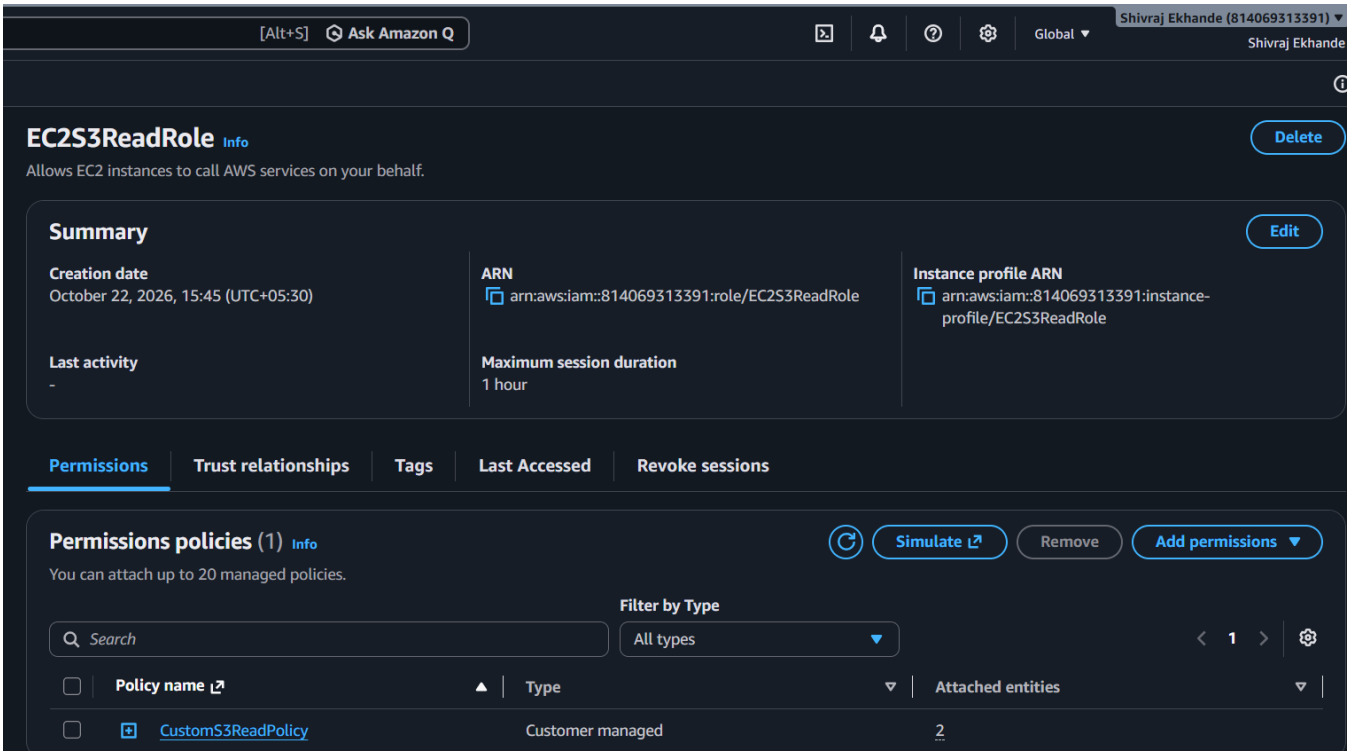

[Alt+S]
Global
Shivraj Ekhande (814069313391)
Shivraj Ekhande
CustomS3ReadPolicy Info
Edit
Delete
Policy details
Type
Customer managed
Creation time
October 22, 2026, 15:30 (UTC+05:30)
Edited time
October 22, 2026, 15:30 (UTC+05:30)
ARN
arn:aws:iam::814069313391:policy/CustomS3ReadPolicy
Permissions
Entities attached
Tags
Policy versions (1)
Last Accessed
Permissions defined in this policy Info
Edit
Summary
JSON
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
(1)
Search
Allow (1 of 477 services)
Show remaining 476 services
Service
Access level
Resource
Request condition
S3
Limited: List, Read
Multiple
None

8. The policy must not grant permission to upload, overwrite, or delete objects.
9. Attach the policy to the appropriate test user or group. Use your actual bucket ARN and object ARN in the policy.

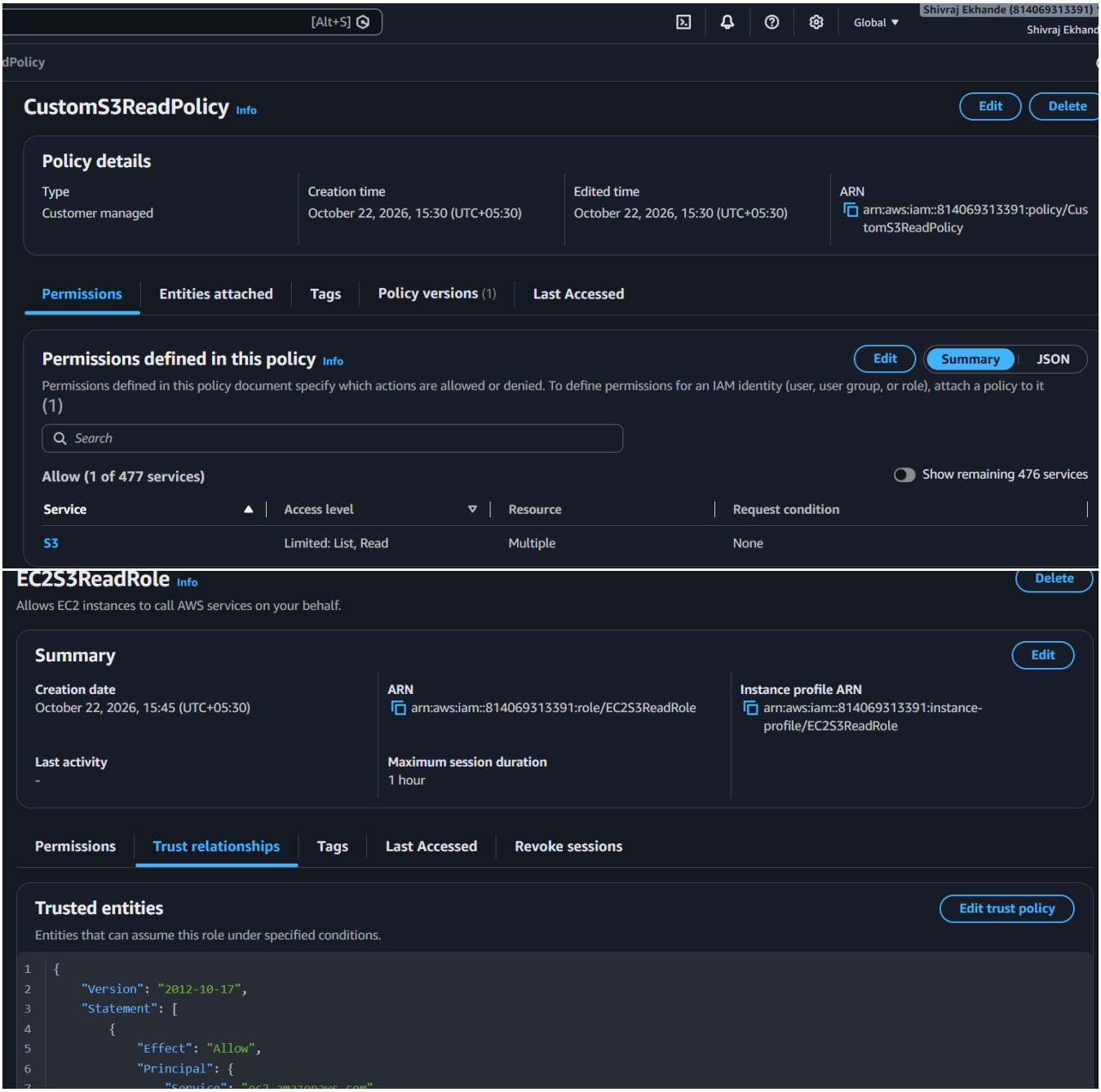


Task 4: Create an IAM role for EC2

10. Create an IAM role named EC2S3ReadRole with EC2 as its trusted service.



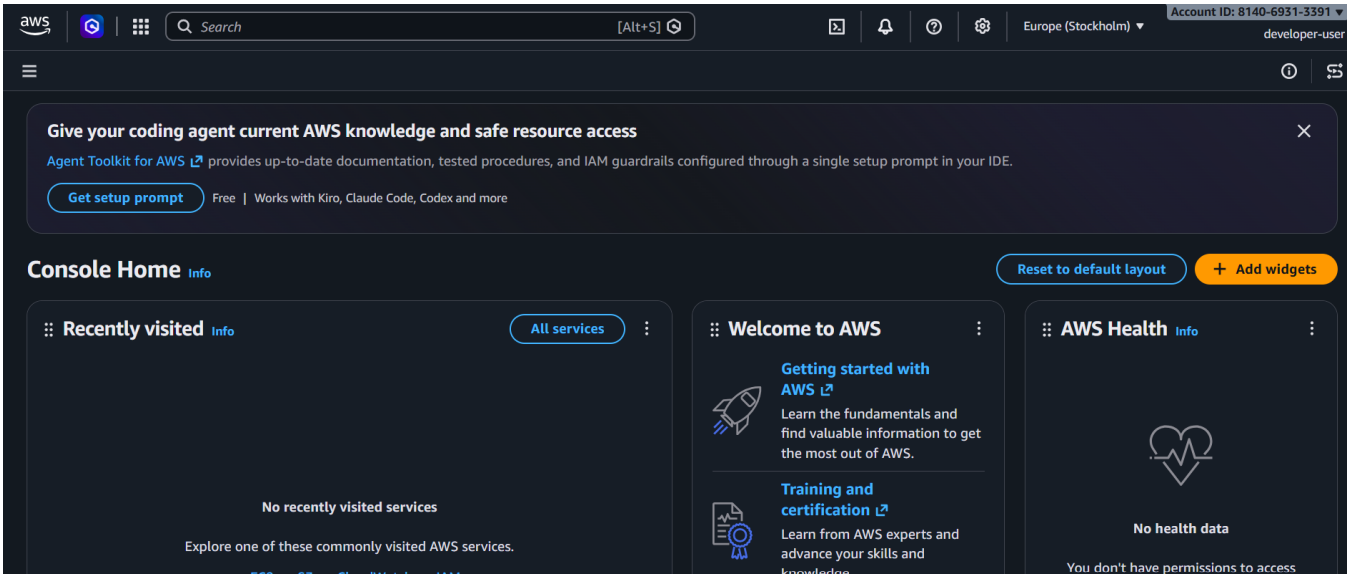
11. Give the role permission to read objects from your practical S3 bucket only.



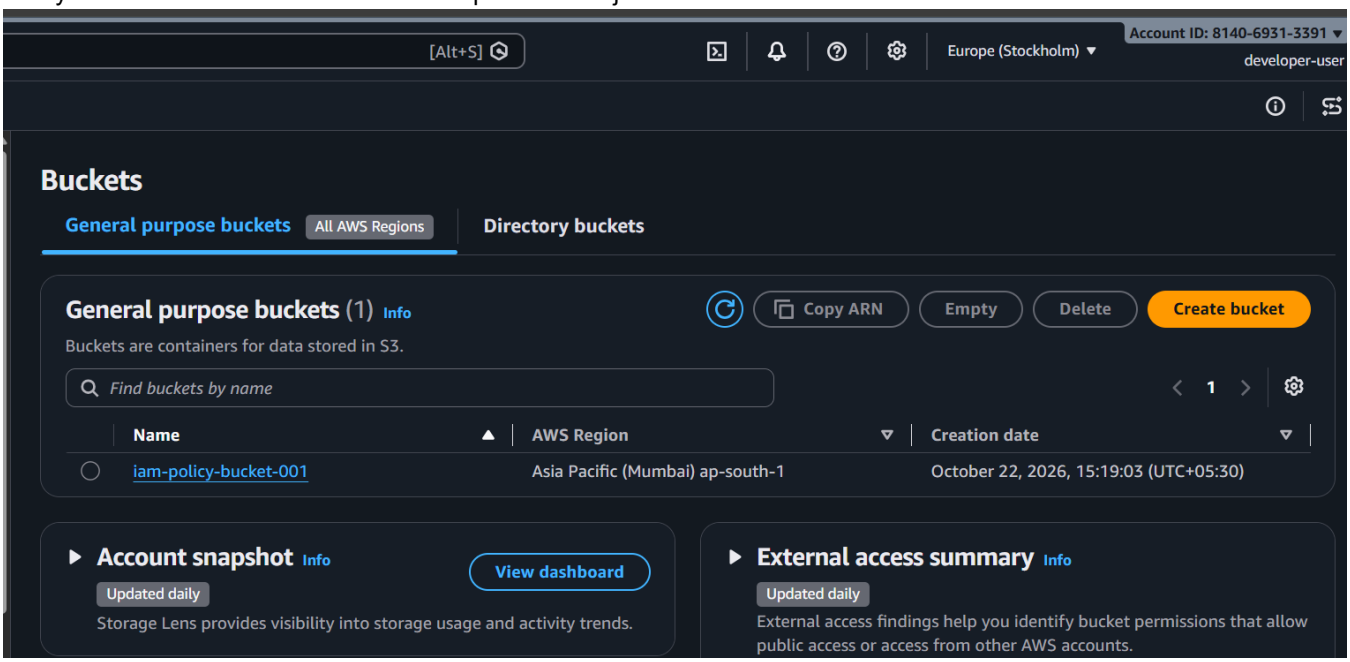
12. Do not create or store long-term access keys for the EC2 instance.
13. You are not required to launch an EC2 instance unless your instructor asks you to.

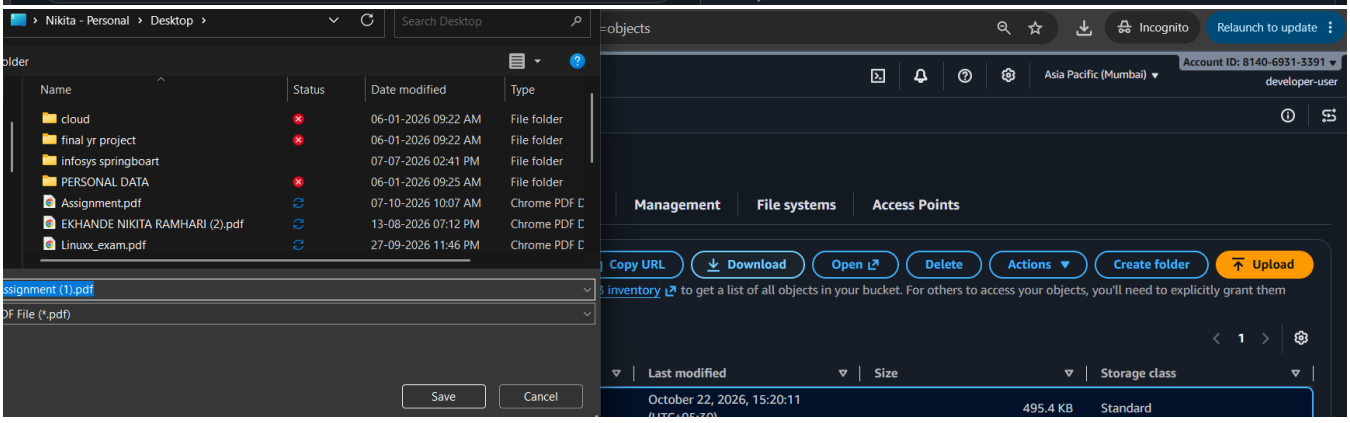
Task 5: Verify IAM permissions

14. Sign in as developer-user or use an instructor-approved test method.

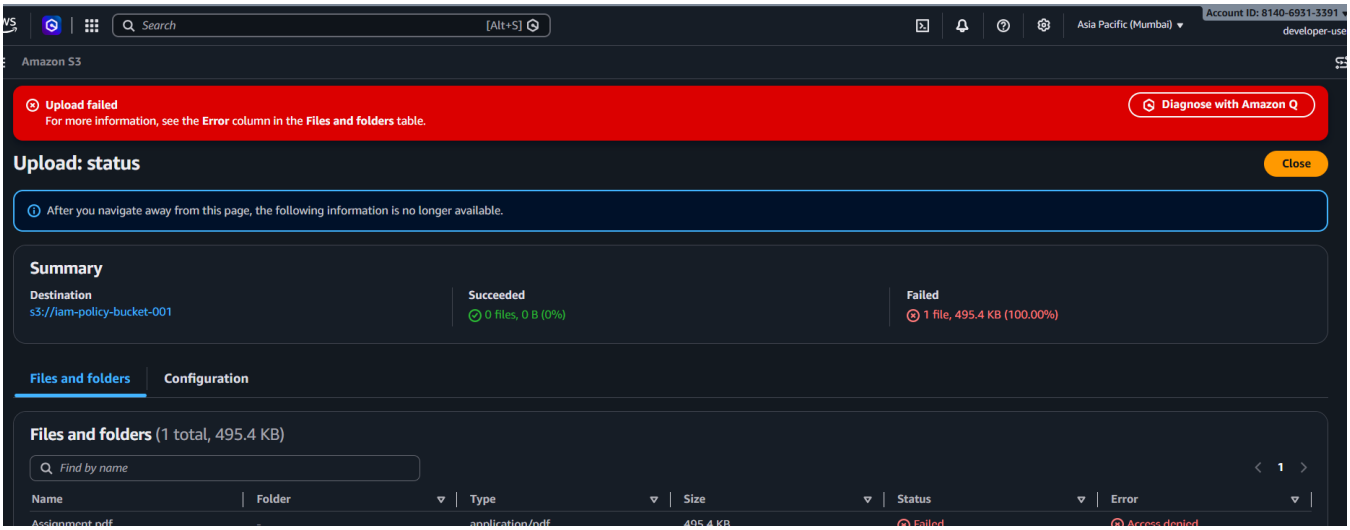


15. Verify that the user can list and download permitted objects.

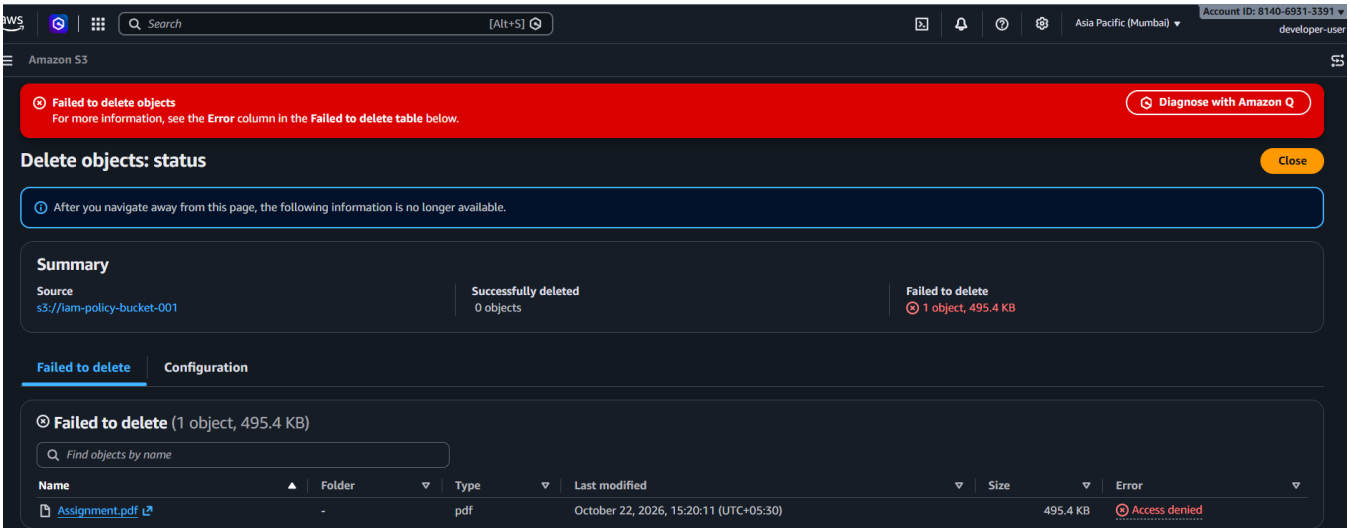




16. Verify that uploading and deleting objects are denied.



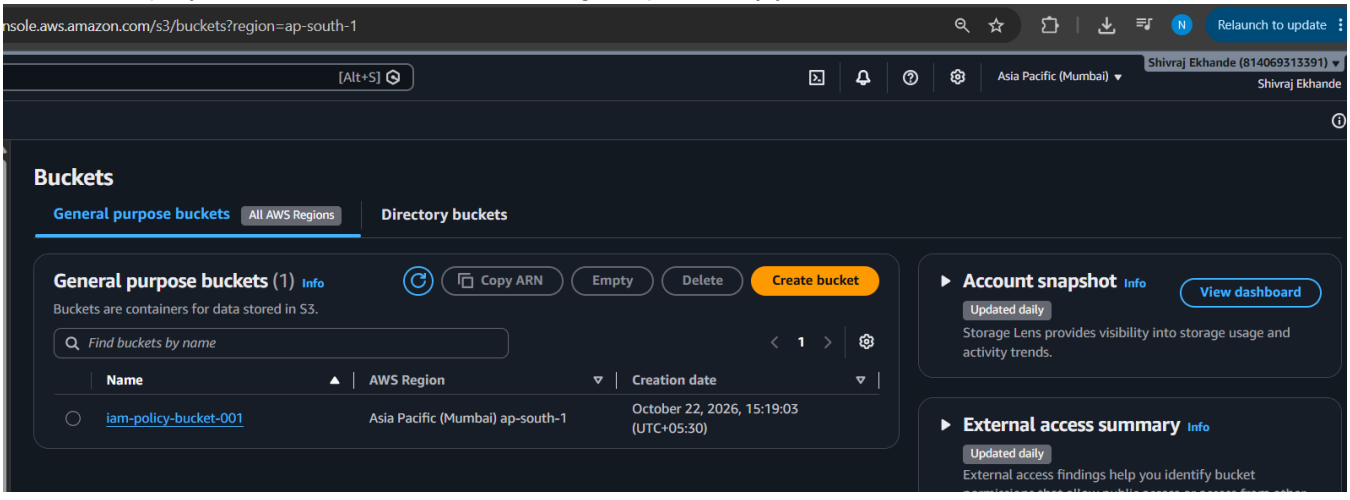
17. Record the action tested and whether it was allowed or denied.



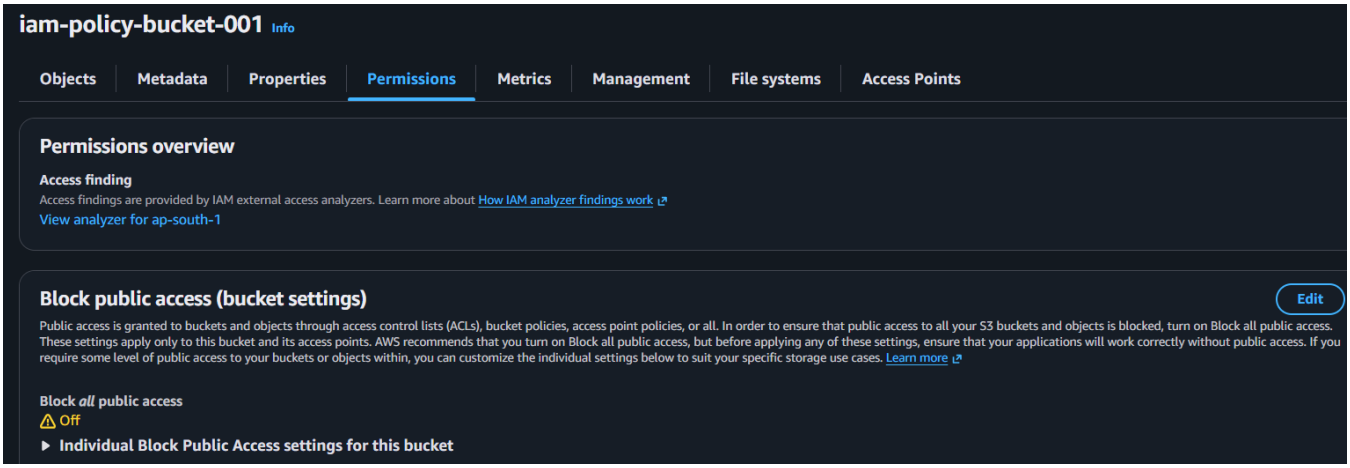
Section B — Amazon Simple Storage Service (S3)

Task 6: Create and secure an S3 bucket

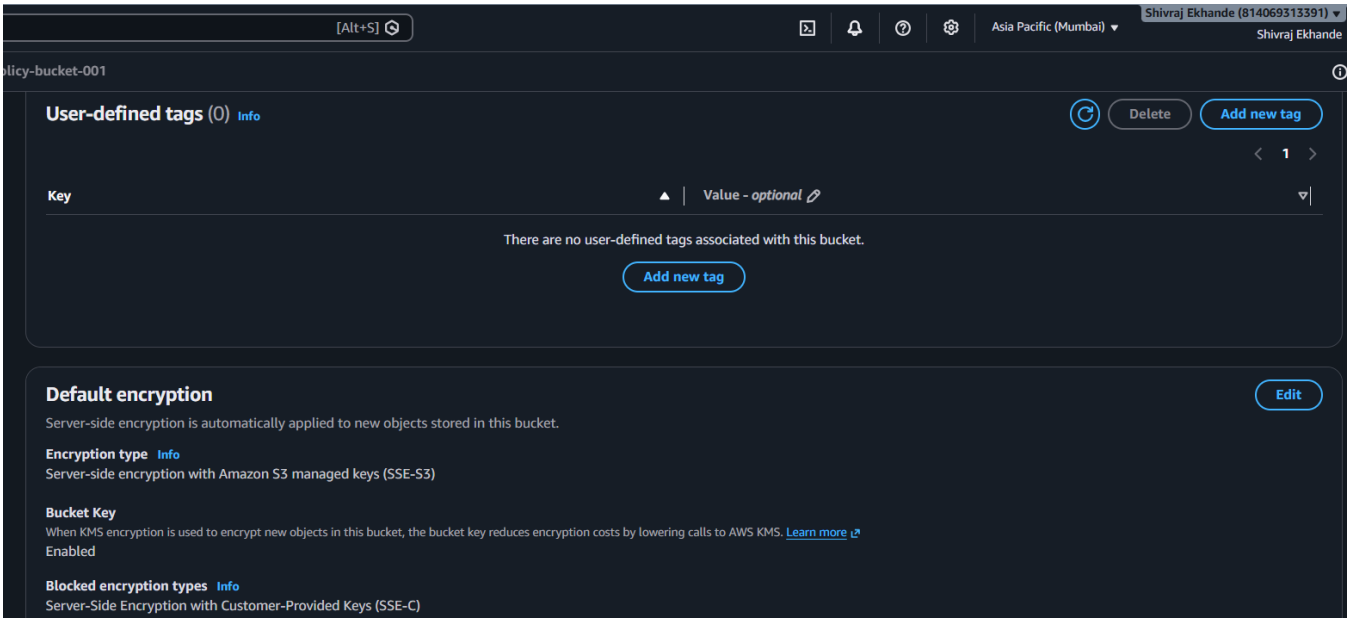
18. Create a uniquely named S3 bucket in the AWS Region specified by your instructor.



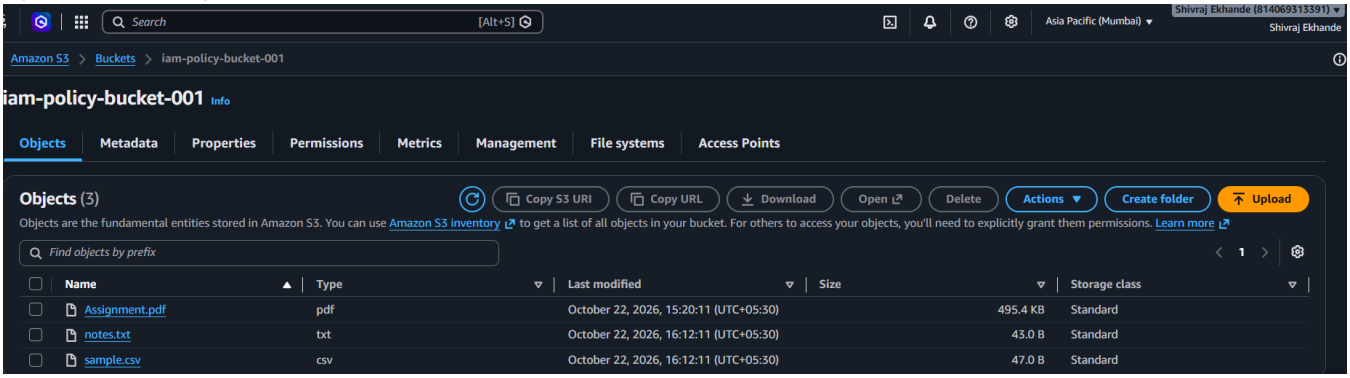
19. Keep Block Public Access enabled.



20. Enable default server-side encryption using the available AWS-managed encryption option.

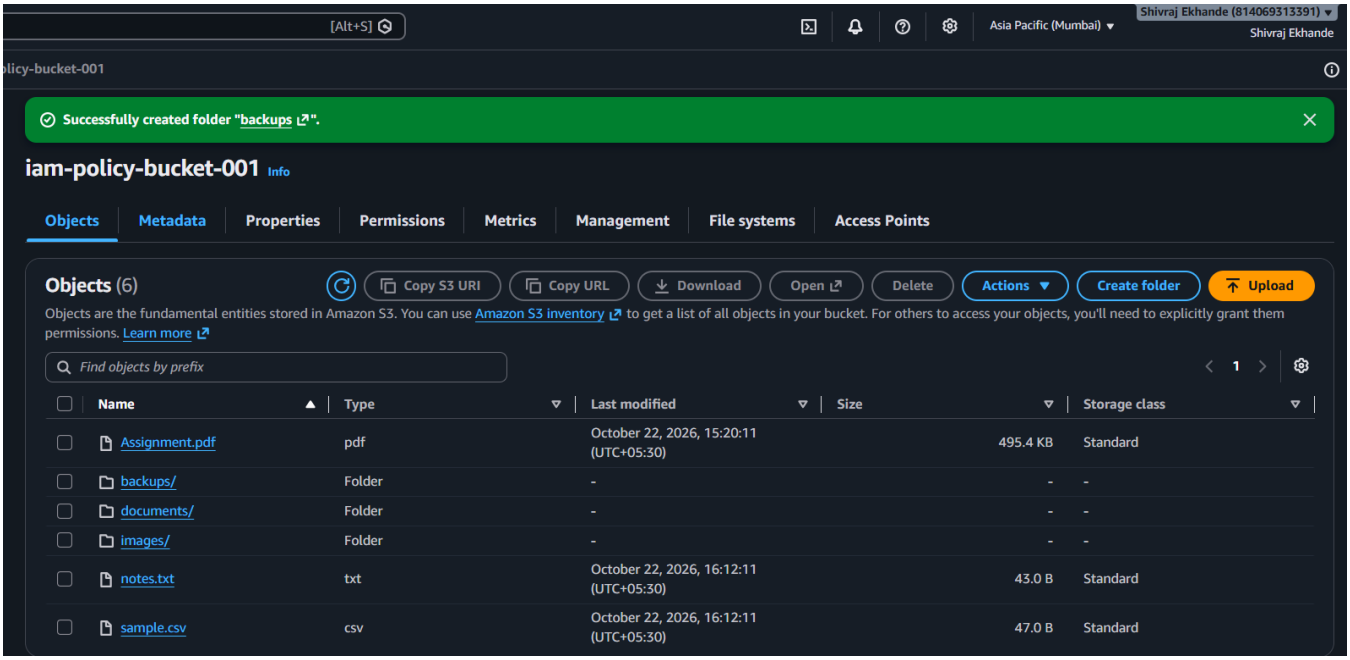


21. Upload three sample files to the bucket.

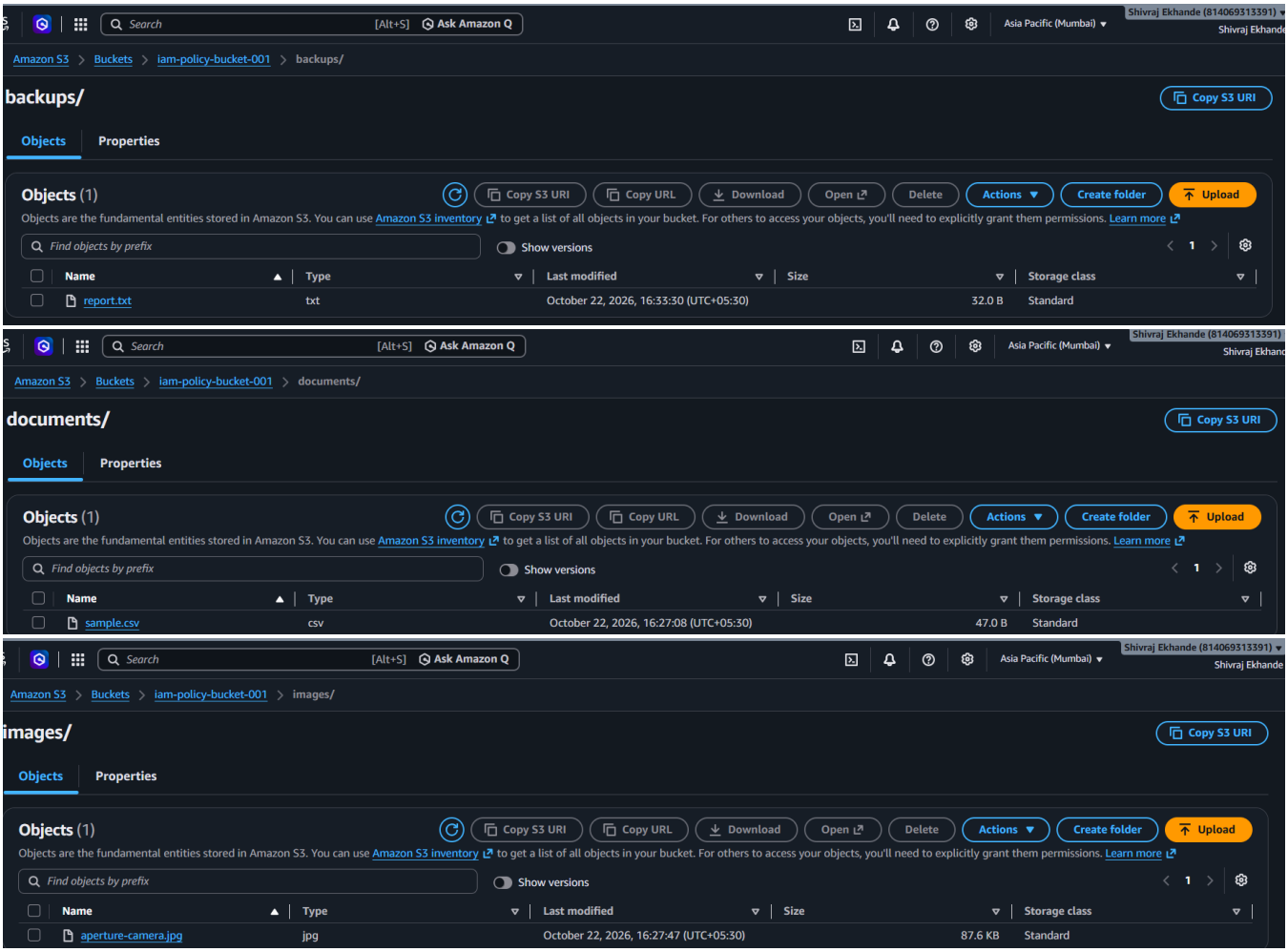


Task 7: Organize objects and enable versioning

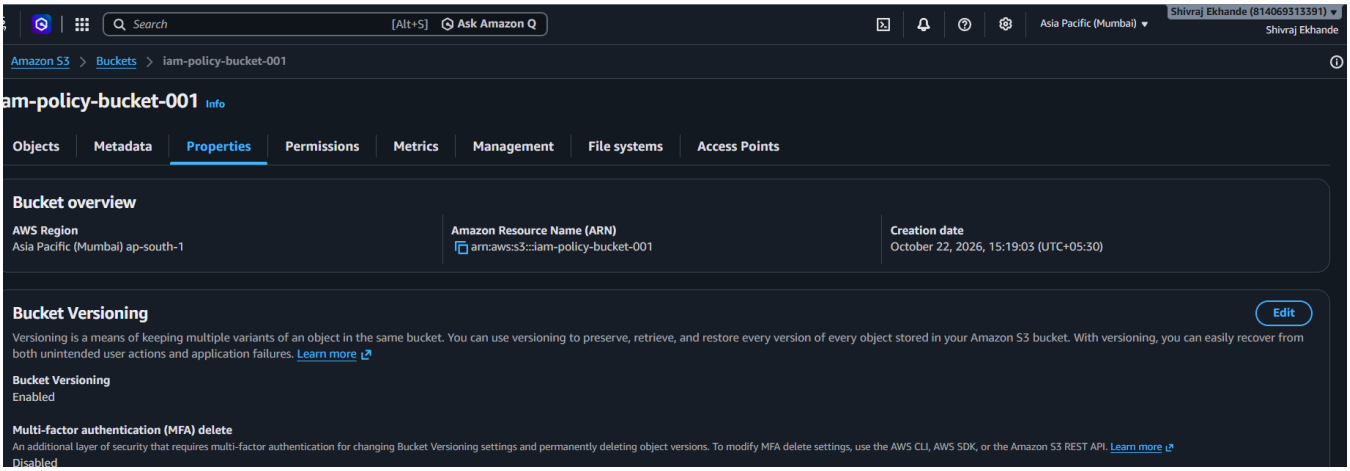
22. Create the prefixes documents/, images/, and backups/ in your bucket.



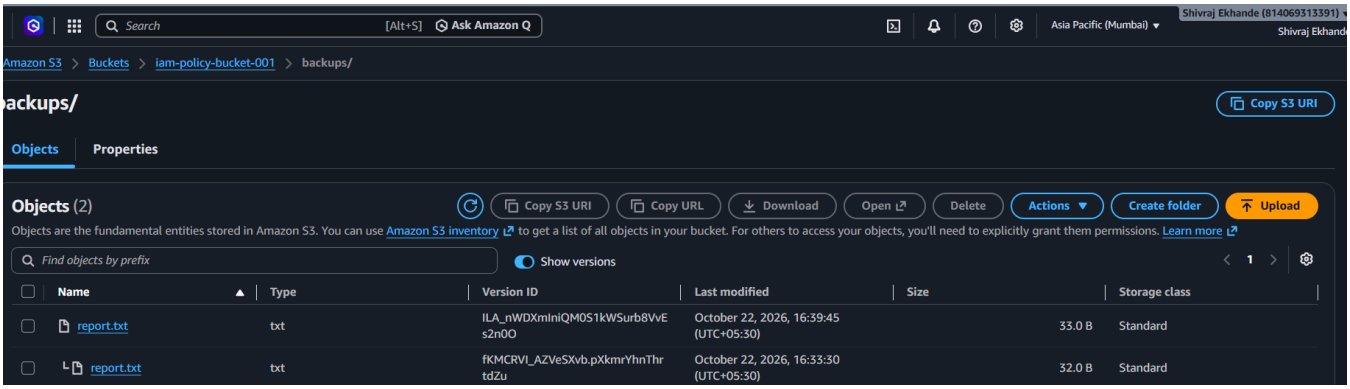
23. Place at least one appropriate sample object under each prefix.



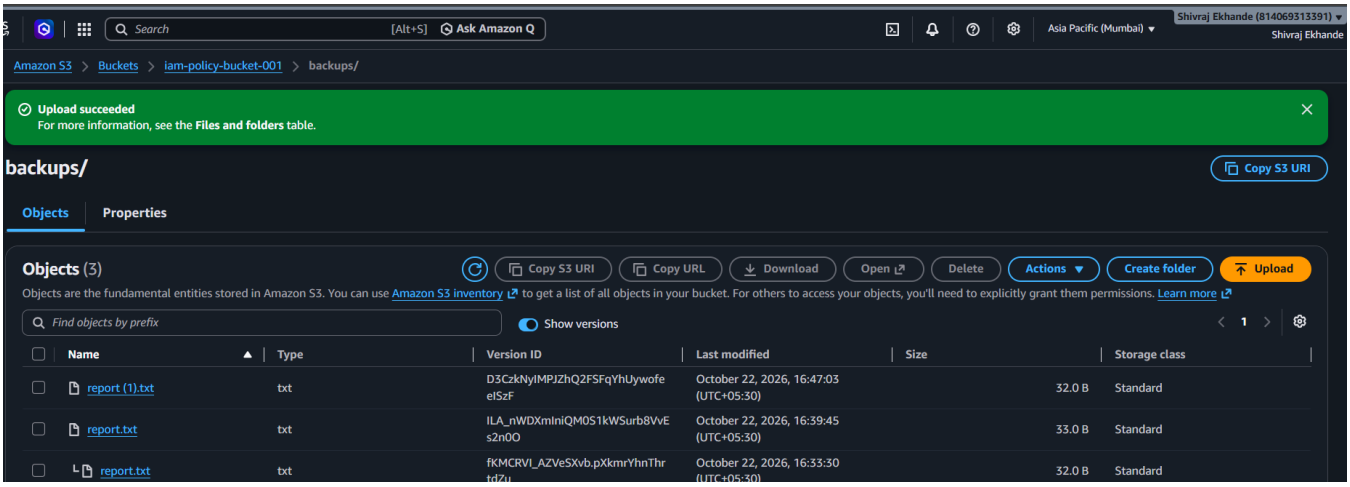
24. Enable bucket versioning.



25. Upload report.txt, change its content, and upload it again using the same object key.

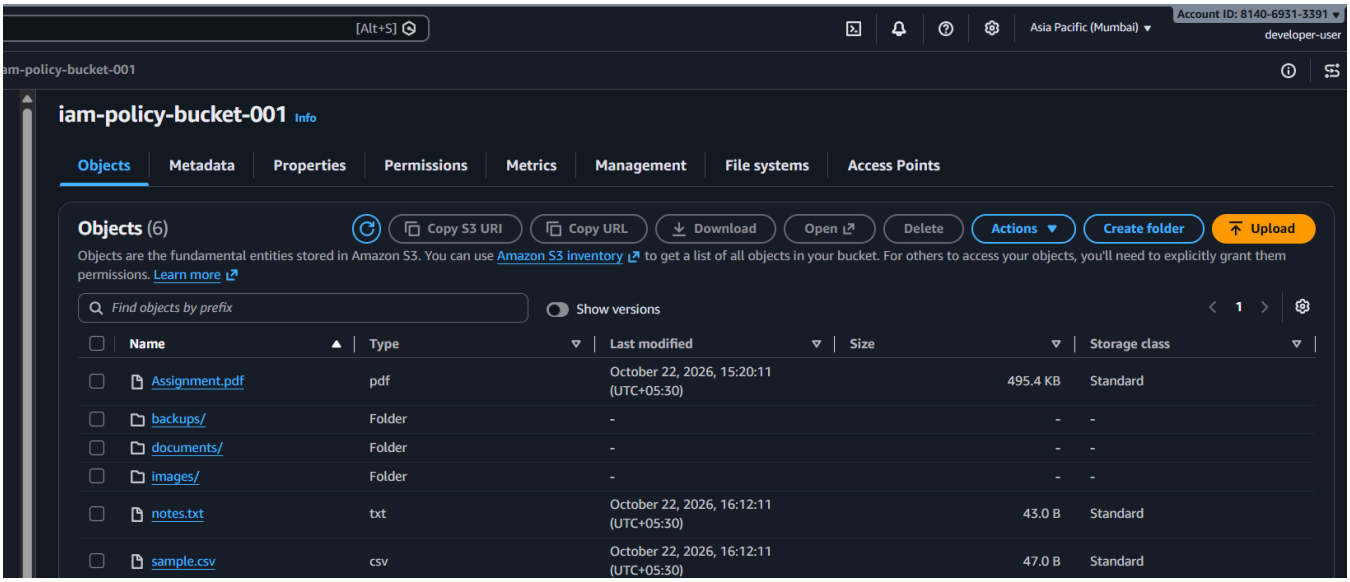


26. Show the versions and demonstrate how to restore the earlier content as the current version.



Task 8: Configure an S3 bucket policy

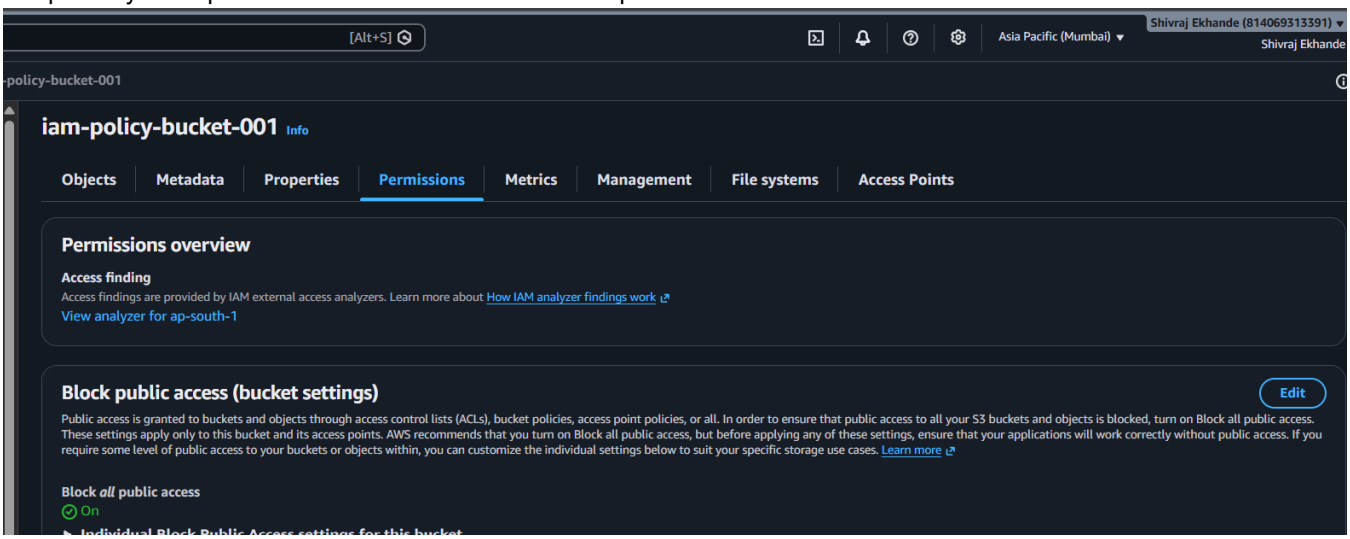
27. Configure the bucket so that the IAM principal developer-user can read objects from it.

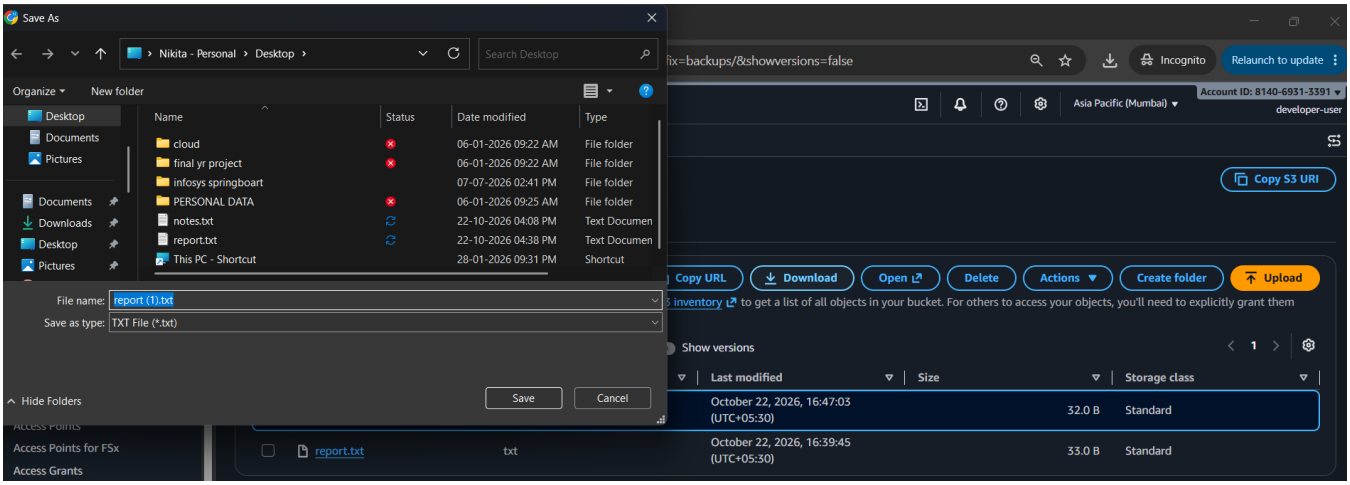


28. Use the correct bucket and object ARNs in the policy.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowDeveloperUserReadObjects",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::814069313391:user/developer-user"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::iam-policy-bucket-001/*"
    }
  ]
}
```

29. Keep anonymous/public access blocked. Do not add a public Allow statement.

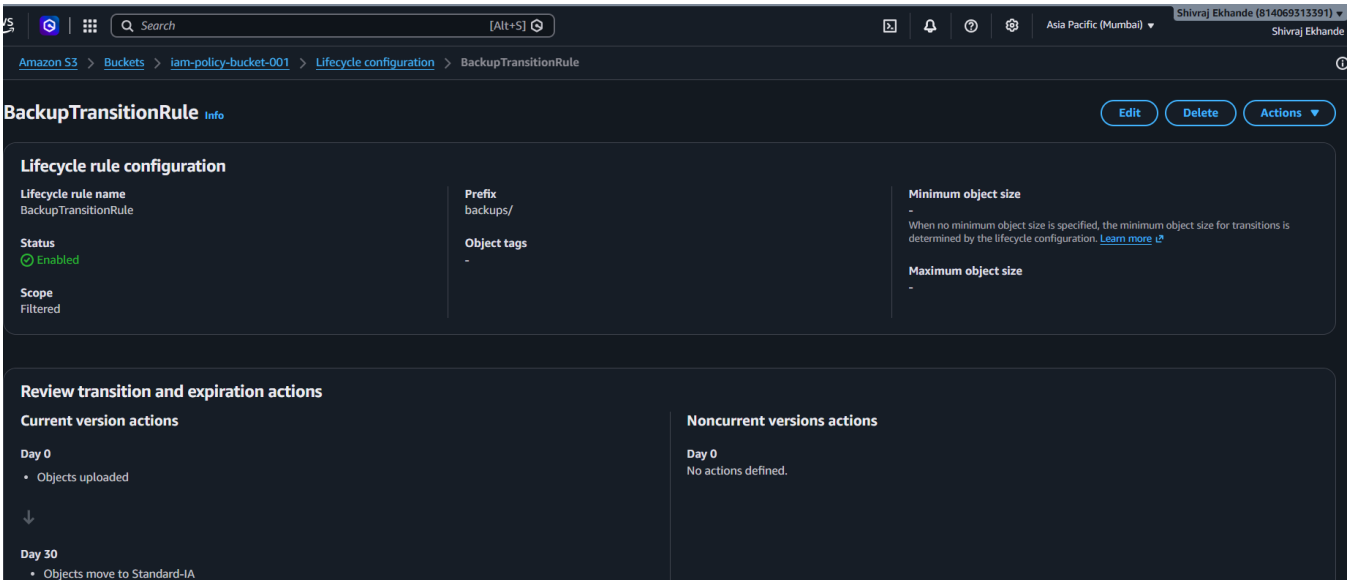




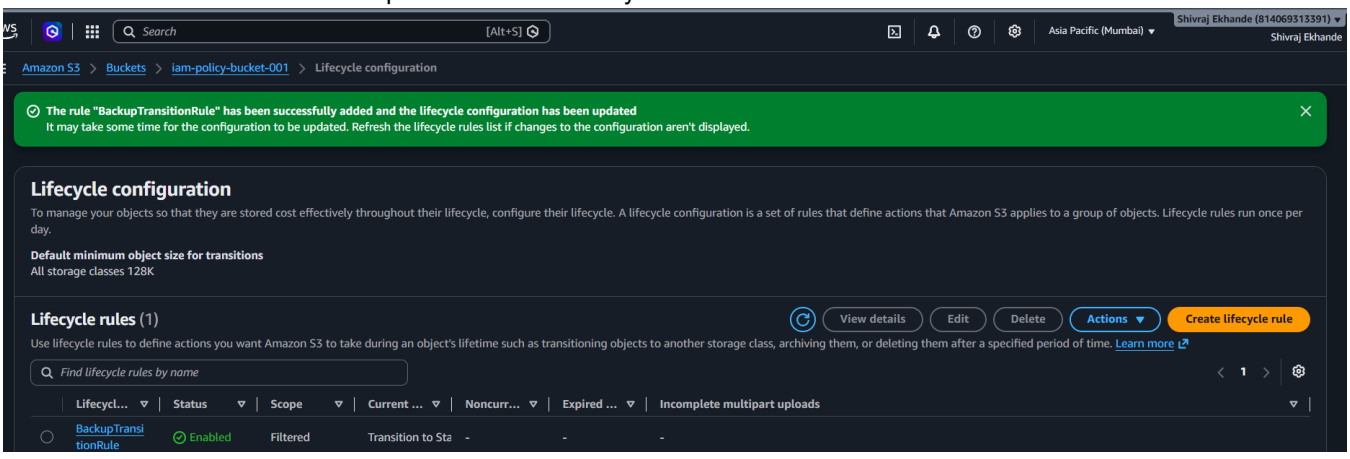
30. Test access as the intended principal and explain how you confirmed that public access remains blocked.

Task 9: Configure a lifecycle rule

31. Create a lifecycle rule named BackupTransitionRule for objects under the backups/ prefix.
32. Configure eligible objects to transition to S3 Standard-IA after 30 days.



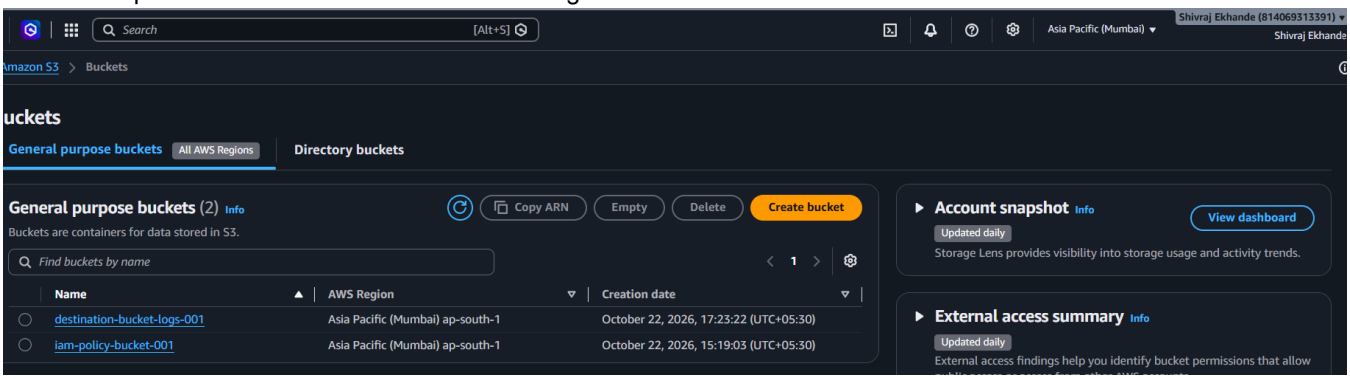
33. Ensure the rule is enabled and inspect the rule summary.



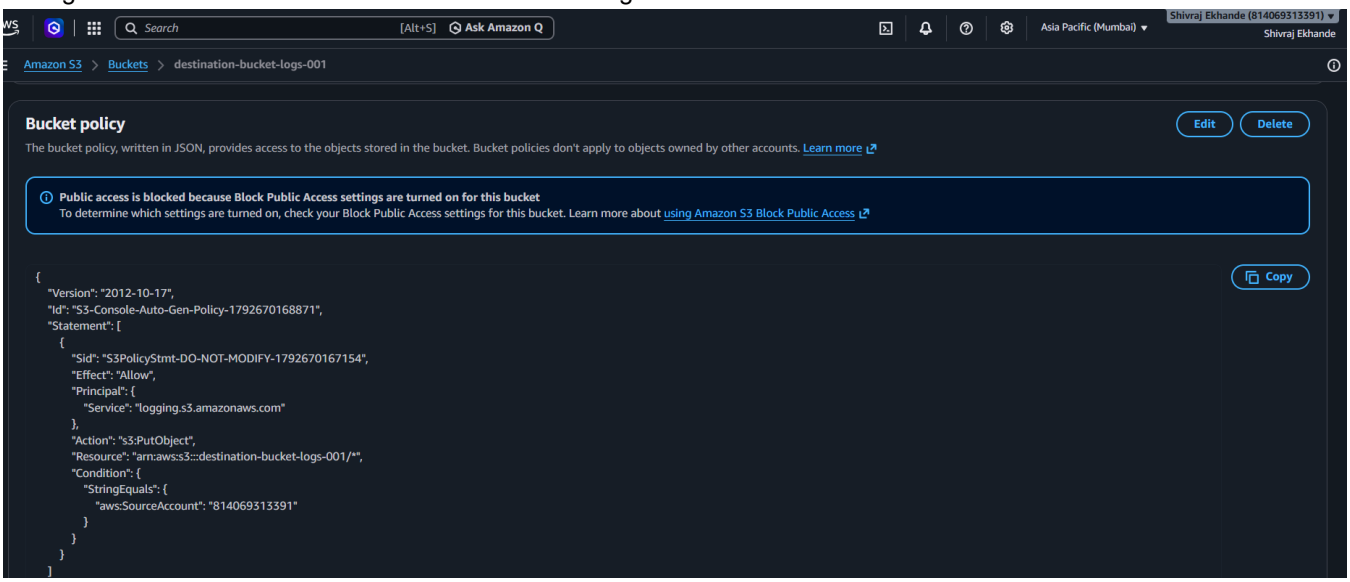
34. Explain why a newly uploaded object may not transition immediately and mention relevant storage-duration or object-size considerations.
    Ans: The lifecycle rule moves eligible objects in the `backups/` folder to S3 Standard-IA after 30 days. New objects may not move immediately because AWS needs time to process the rule. S3 Standard-IA has a minimum storage duration of 30 days.

Task 10: Configure server access logging

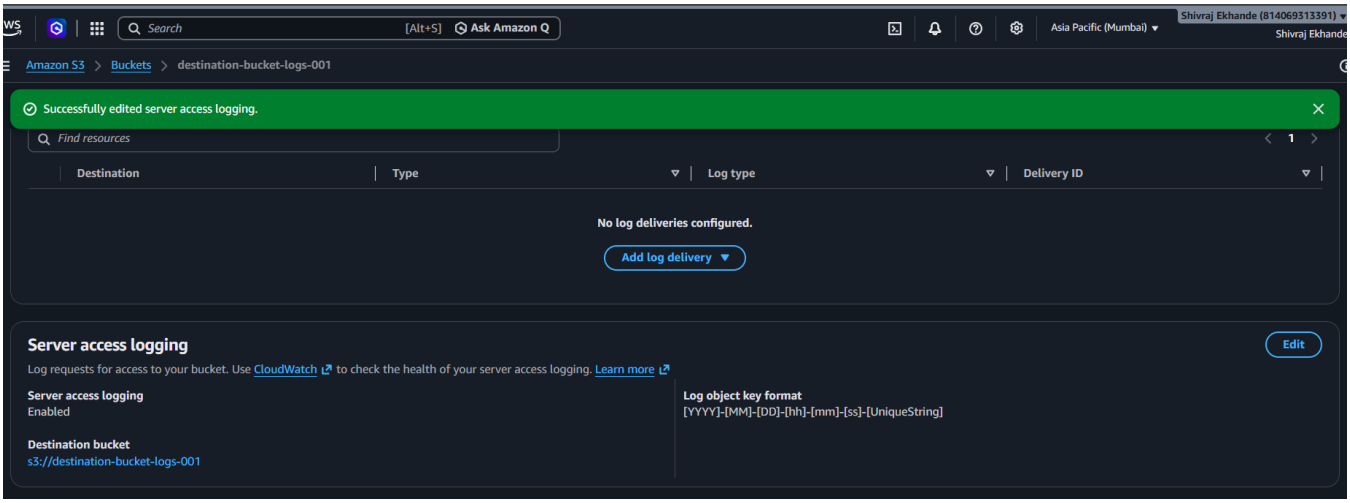
35. Create a separate destination bucket for access logs.



36. Configure the source bucket to deliver server access logs to the destination bucket.



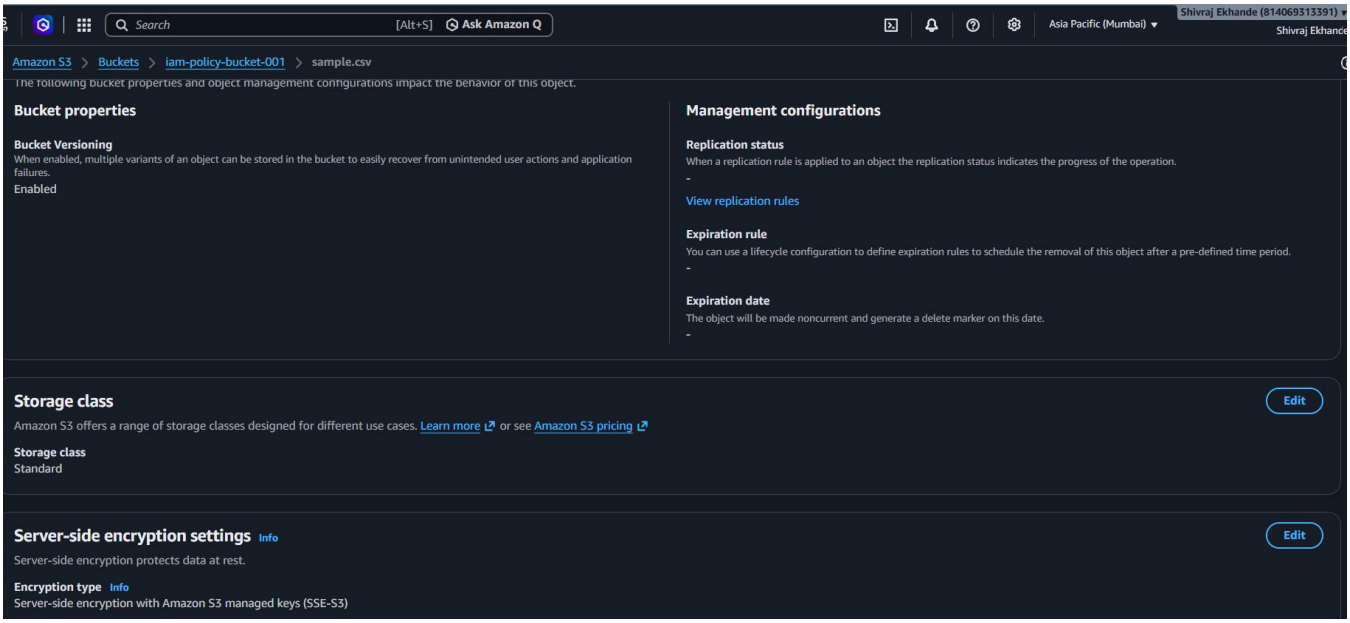
37. Verify the logging configuration and destination permissions.



38. State why logs may not appear immediately after you enable logging.

Task 11: Explain encryption and access controls

39. Open the properties of one uploaded object and identify the server-side encryption setting.



40. Explain the difference between IAM permissions and an S3 bucket policy.
    IAM Policy: Defines what actions an IAM user or group can perform on AWS resources.
    S3 Bucket Policy: Defines who can access an S3 bucket and its objects, and what actions they can perform.

41. Explain why Block Public Access is useful.
    Block Public Access helps prevent accidental public access to S3 buckets and objects. It protects data from being accessed by unauthorized users.